Project 2: Internal Control and Firm Performance

Background

In this project, your group will act as a team of managers debating whether (and how much) to invest in improving your internal controls over financial reporting (ICFR).

Internal control is a process designed to provide reasonable assurance regarding a firm's objectives in operational effectiveness and efficiency, reliable financial reporting, and compliance with laws, regulations, and policies. ICFR is a subset that primarily focuses on financial reporting.

After the Enron scandal, Section 404 of the Sarbanes-Oxley Act (SOX 404) requires public firms to establish, maintain, assess, and report on the effectiveness of their ICFR. The primary objective is to ensure that companies have effective controls to prevent or detect material misstatements in their financial statements. In particular, SOX 404(a) requires management to assess the effectiveness of ICFR and disclose any identified material weaknesses, and SOX 404(b) requires auditors to attest to the effectiveness of ICFR.

In your prior accounting classes, you (may) have learned that good ICFR is important, and that one way it helps is by supporting operations. For example, high-quality ICFR ensures information accuracy. Accurate information can facilitate managerial decision-making and, hence, improve firm performance. Thus, while improving ICFR can be costly, improved firm performance can "offset" these costs or even lead to net gains.

While this argument makes theoretical sense, we do not know whether it is supported by real-world data. In addition, to make a more informed decision on how much to invest, your group needs to quantify the benefit of high-quality ICFR.

We will measure the quality of ICFR using internal control material weaknesses (ICMWs), which are severe flaws in a company's ICFR that create a reasonable possibility that a material misstatement of the company's financial statements will not be prevented or detected in a timely manner.

ICMWs can arise from many areas. Your CIO suggests that investing in better IT can help address IT-related ICMWs. However, you are unsure whether you should prioritize that over non-IT-related ICMWs such as those from employee training. In a later analysis, we will distinguish IT-related ICMWs from non-IT-related ICMWs.

The instructions below will guide you through this analysis. As usual, some instructions are written "vaguely" to allow for flexibility.

Tools

We will primarily use Python. It is also okay to use other applications, such as Excel, for small tasks.

Regressions

We will estimate two regressions; the first one is as follows.

$$ROA = \beta_0 + \beta_1 ICMW + \beta_2 \ln(assets) + \beta_3 leverage + \beta_4 BTM + \beta_5 receivables + \beta_6 inventory + \beta_7 big4 + \varepsilon$$

where

- *ROA* represents return on assets (detailed variable definitions are provided in the next section),
- *ICMW* is an indicator for a firm suffering from internal control material weaknesses,
- *assets* represents total assets, and we will take log,
- *leverage* is the ratio of debt to assets,
- *BTM is book to market ratio,*
- *receivables* is the ratio of receivables to total assets,
- *inventory* is the ratio of inventories to total assets,
- *big4* is a binary indicator for the firm's auditor type (*big4* = 1 means Big 4 auditor, and = 0 means non-Big 4 auditor).

The second regression examines IT-related ICMWs (ITMW) and non-IT-related ICMWs (nITMW) separately as follows.

$$ROA = \beta_0 + \beta_{1a} ITMW + \beta_{1b} nITMW + \beta_2 \ln(assets) + \beta_3 leverage + \beta_4 BTM + \beta_5 receivables + \beta_6 inventory + \beta_7 big4 + \varepsilon$$

To estimate the regressions above, you will need data from Compustat (North America, Annual) and Audit Analytics. Because you will not use CRSP data, you can use the "original" Compustat from WRDS, not the CRSP/Compustat merged file.

Regression variables are defined as follows. As you can see, we use lagged values in many variables (in bold font below). This is because lagged values are essentially the beginning-of-year values. Thus, they are "cleanner" as they are unaffected by concurrent ROA. That is, we want the variables such as total assets to determine ROA, but not ROA determine total assets. This is a common practice in accounting and finance research.

- ROA: Compustat net income (data item: ni) divided by **lagged** total assets (data item: at).

- ICMW, ITMW, nITMW: These are internal control weakness measures. They are defined below in Step 2 when we describe the internal control data (page 4).

- assets: **lagged** Compustat total assets (data item: at). Remember to take log.

- leverage: Compustat provides short-term debt (data item: dlc) and long-term debt (data item: dltt). We should set missing values to 0, and leverage is (**lagged** dlc+ **lagged** dltt) divided by **lagged** at.

- BTM: Compustat provides book value of common equity (data item: ceq), while market value of common equity is end-of-year price (data item: prcc_f) times common shares outstanding (data item: csho). BTM is **lagged** ceq divided by (**lagged** prcc_f times **lagged** csho).

- receivables: Compustat **lagged** total receivables (data item: rect) divided by **lagged** total assets (data item: at).

- inventory: Compustat **lagged** total inventories (data item: invt) divided by **lagged** total assets (data item: at).

- big4: Compustat (data item: au) provides auditor information, which is a number used to indicate different auditors (e.g., 6 is KPMG, 7 is PwC, and 17 is Grant Thornton). A value <= 8 indicates Big 4 auditors. We should create a binary indicator that = 1 for Big 4 auditors, and 0 for non-Big 4 auditors. **This variable doesn't need to be lagged** because we want to examine current auditors, not prior-year auditors.

Steps

When you complete these steps, you will also create a report (**a Word document**) describing what you find. Report requirements are included below.

Step 1. Download Compustat Data.

In WRDS, download Compustat (North America -> Fundamental Annual) data by setting the date range to Jan 2010 - Dec 2025 (datadate).

We will **search the entire database** to download the data items mentioned above, as well as the following data items:

- cik: Central Index Key, which is a firm identifier created by the U.S. Securities and Exchange Commission (SEC).
- sic: SIC code for industries.
- datadate: The ending date of a fiscal year. **This data item should be automatically included in your download, so no need to select it from the list.**

Note on report: Report the number of rows you have downloaded (excluding the header row).

Step 2. Download Audit Analytics – Internal Control Data.

In WRDS, find "Audit Analytics." Scroll down to find "Audit + Compliance" -> "SOX 404 Internal Controls".

"Step 1" in Audit Analytics allows you to choose a file date (the date a 10-K was submitted to the SEC). Because the file date is after the fiscal year end, we will set the date range to end slightly after Dec 2025, for example, May 31, 2026. For the starting date, we will set it to Jan 1, 2011. This is 1 year after the start date of our Compustat data. Why? Because we need to use many lagged values from Compustat but not from Audit Analytics. Again, search the entire database.

Then download the following data items:

- fye_ic_op: the ending date of a fiscal year (similar to datadate in Compustat).
- ic_op_type: the type of internal control opinion; the value could be "m" or "a" indicating whether a row contains the managers' or auditors' assessment. That is, for each firm-year combination, there could be two records, one for manager (m) and one for auditor (a). **We want to focus on managers' assessment (i.e., type "m")**.
- ic_is_effective: an indicator for whether the assessment shows internal control is effective or not. The value can be "Y" (i.e., effective), "N" (i.e., not effective), and rarely, other values such as "ND" (i.e., not disclosed). **Let's drop any value other than "Y" and "N."**
- noteff_other_reas_keys: this column tells you reasons why internal control was considered ineffective. The reasons are coded. For example, you might see values such as "|5|9|17|42|44|." Here, 5, 9, 17, 42, 44 are 5 different reasons for internal control to be inefficient. Let's focus on code 22, which indicates IT-related weaknesses.

By default, your download will also include company_fkey, which is equivalent to cik.

Now, in the regressions, we need to create three internal control measures: ICMW, ITMW, and nITMW. They are defined as follows:

- ICMW: this is an indicator that equals 1 if a firm suffers from internal control material weaknesses (i.e., ic_is_effecitve is "N"), and 0 otherwise.
- ITMW: this is an indicator that equals 1 if a firm suffers from IT-related internal control material weaknesses (i.e., noteff_other_reas_keys contains "22"), and 0 otherwise.
- nITMW: this is an indicator that equals 1 if a firm suffers from ICMW but not ITMW (i.e., ICMW = 1 and ITMW = 0), and 0 otherwise.

Please note that we will use the concurrent ICMW/ITMW/nITMW measures. That is, no need to use the lagged values.

Note on report: Report the number of rows you have downloaded (excluding the header row).

Step 3. Data Cleaning.

Now, let's switch to Python and start data cleaning. Open Jupyter Notebook and create a new Notebook file (in the same folder as the data files; If asked, select Kernel "Python 3"). Use

Markdown or comments to briefly describe each data-cleaning step in the file. You will submit this file along with your report.

The prepared data after this step should contain all the variables needed for regression analysis. You have seen many data cleaning examples before, so this instruction doesn't contain specific steps for you to follow (except for those discussed in Step 2). Instead, please use your best judgment, and the following additional conditions are suggested:

- at, prcc_f, csho, ceq, au must be positive.
- ni and cik must be non-missing.
- Replace missing dlc, dltt, rect, invt with 0.
- Remove financial service firms.
- After the previous steps, remove all records with missing values for the variables used in the regression (e.g., ROA, leverage, etc.).

I recommend first cleaning each dataset (but not winsorizing), joining the datasets, and winsorizing should be the last step before regression analysis**. Winsorization is for continuous variables only.** To join Compustat with Audit Analytics, we can use the following columns:

- cik and company_fkey are essentially the same attribute.
- datadate and fye_ic_op are both the ending date of a fiscal year.
- It's recommended to ensure that the data types are consistent for the matched attributes.

Note on report: In the Word report, explain your data cleaning process. Please be clear but concise. In addition, please report, after cleaning, (1) what is the number of observations used for regression analysis, (2) what % of observations suffer from ICMW, and (3) what % of observations suffer from ITMW.

Step 4. Linear Regression.

Estimate the two regressions above. For the first regression, check if the coefficient on ICMW is positive or negative, and whether it is statistically significant.

For the second regression, check if the coefficients on ITMW and nITMW are positive or negative, and whether they are statistically significant.

In addition, we want to test whether the coefficients on ITMW and nITMW differ significantly. The coefficients are estimated from a sample, and, as in our discussion of hypothesis testing, they are random (e.g., if we change the sample, the coefficients can change). In other words, if we find the magnitude of the coefficients is different, it does not necessarily mean that the coefficients are different statistically (just like for hypothesis testing, a difference in the means doesn't necessarily imply that the means are statistically different).

We will use the following Python code to test (some modification might be needed – AI can be very helpful!):

```
constraint = "ITMW - nITMW = 0"
t_test_res = regression2.t_test(constraint)
print(t_test_res)
```

A small p-value indicates that the coefficients are statistically different.

Note on report: In the Word report, attach screenshots of the linear regression results and coefficient comparison results. Discuss your findings.

Step 5. Cost-Benefit Analysis.

Next, analyze how much you, as managers, will invest in internal controls to improve firm performance. Currently, your firm suffers from ICMWs, and correcting the problems is costly. However, if resolving the problem improves ROA, the increased profitability can offset the costs.

In particular, please consider the following cost-benefit analysis. The cost is additional investments/resources needed to ensure no ICMWs. The benefit comes from improved ROA. Assume the total assets of your firm is $ 100 (million).

The investment in internal control makes sense if the cost is lower than the benefit. If the cost is $1 (million), should you invest? What is the maximum cost that makes economic sense to your firm? Make additional assumptions as needed.

Also, based on your ITMW and nITMW comparison, should you follow your CIO's suggestion and prioritize addressing IT-related weaknesses?

Step 6. Limitations.

Lastly, please discuss the limitations of your analysis. For example, this analysis makes important explicit and implicit assumptions, some of which are not necessarily that realistic. In addition, other aspects may have been omitted from this analysis, but they could still be relevant.

Your submission should include:

1. The Word report.
2. HTML file exported from Jupyter Notebook.